higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION
(First Paper)
NQF LEVEL 2

12 MARCH 2019

This marking guideline consists of 5 pages.

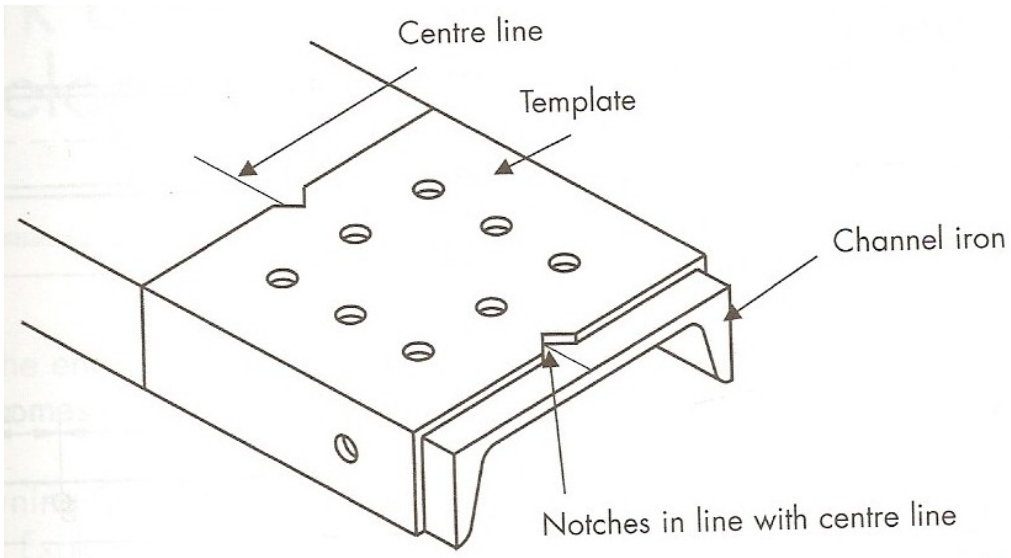
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QUESTION 1

1.1 A template is a device made up of a reliable flat sheet of material with a purpose of copying, while a pattern refers to a material that has been cut to exact size and shape to produce the required product. (2 + 2) (4)

1.2 True (1)

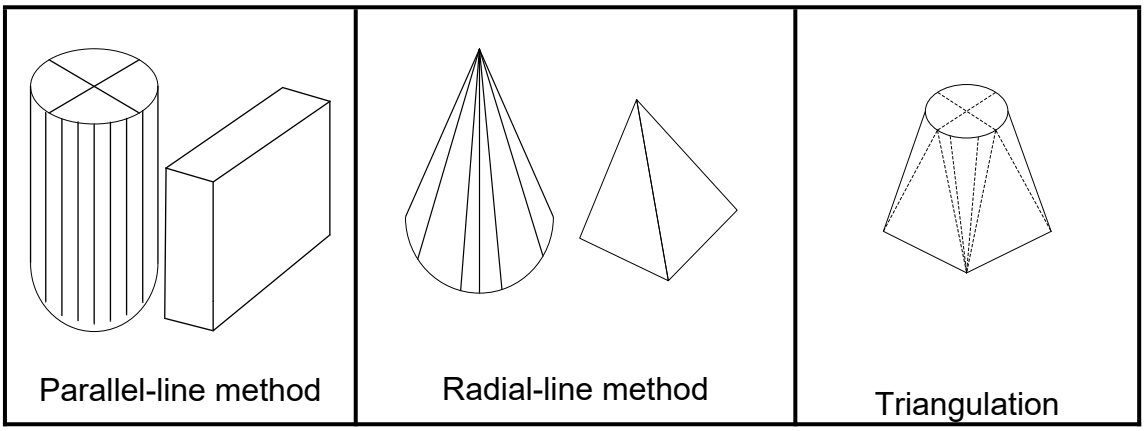
1.3



(Any relevant sketch) (4)

1.4
- Parallel-line method
- Radial-line method
- Triangulation method (3)

1.5



(3)

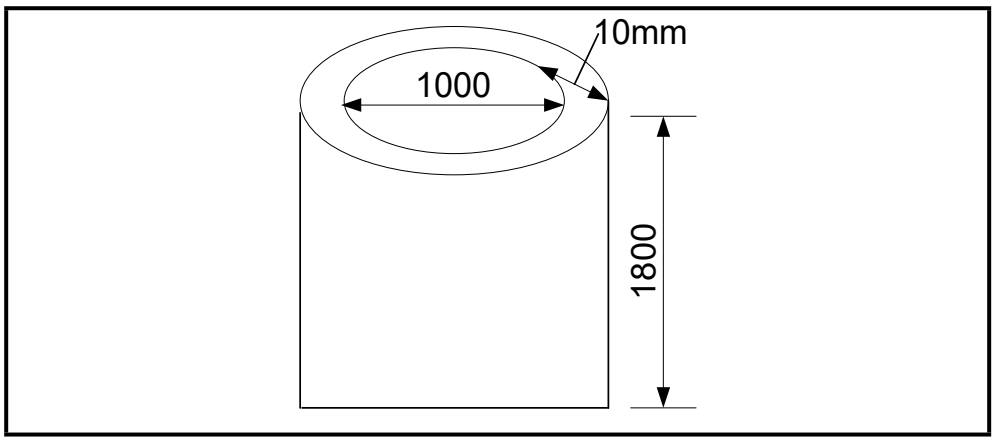
[15]

QUESTION 2

2.1 Mean diameter refers to a diameter formed by the mean line, while inside diameter refers to a diameter formed by the contraction line. (2 + 2) (4)

2.2
- It allows a metal to be rolled.
- It allows a metal to be bent. (2)

2.3 2.3.1



(4)

2.3.2 MD = 1 000 + 10
= 1 010 mm (1)

2.3.3 Area = C × H

But A = MD × 3,142

= 1 010 × 3,142

= 3 173,42 mm

Area = 3,173 × 1,8

= 5,711 m^2 (3)

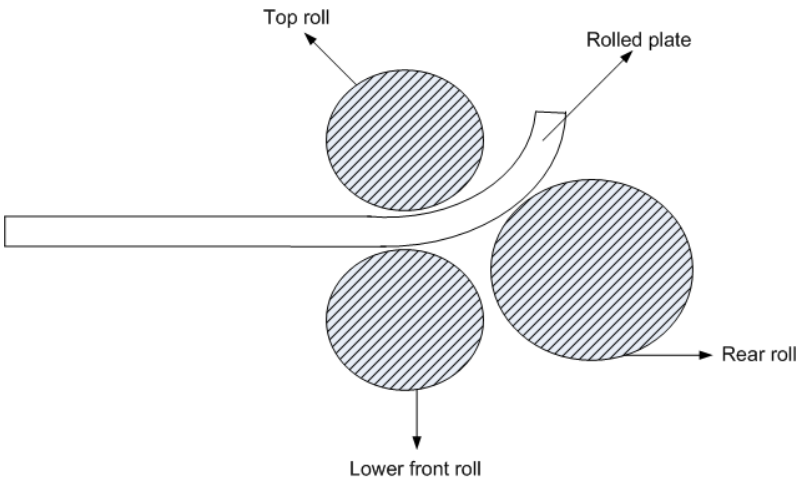
2.4 False (1)
[15]

QUESTION 3

3.1 3.1.1
- Bar folder
- Cornice brake
- Box and pan brake (3)

3.1.2 Bending machines are mainly used to bend sheet metal plates and steel plates whereas forming machines are mainly used for rolling cylinders. (2 + 2) (4)

3.2



(2 marks for presetting) (6)

3.3 3.3.1 True
3.3.2 False

(2 × 1) (2)
[15]

QUESTION 4

4.1 4.1.1
- Oxygen
- Acetylene

(2)

4.1.2
A Oxygen
B Acetylene
C Black/Green
D Left-handed

(4 × 1) (4)

4.2 Carburising flame: excess acetylene



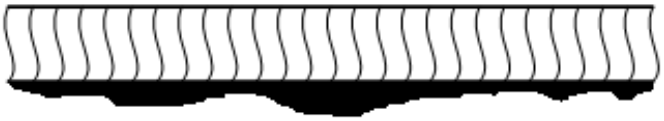
Oxidising flame: excess oxygen



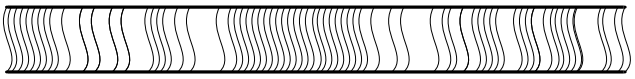
(2 + 2) (4)

4.3 False (1)

4.4 4.4.1 Nozzle held too high:



4.4.2 Nozzle held too close:



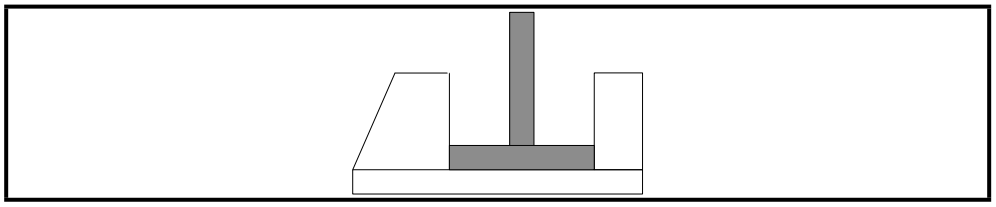
(2 × 2) (4)
[15]

QUESTION 5

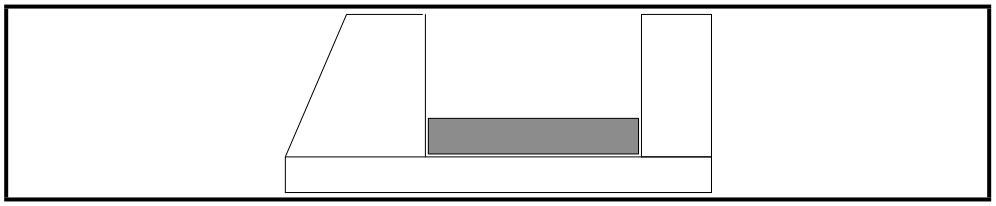
5.1
- Higher cutting speeds and feeds are possible.
- The life of the blade is extended.
- Power consumption is reduced.
- Surface finish and dimensions are improved.
- Swarf is carried away from the work area. (5)

5.2 An abrasive cutting saw uses a cutting disc and no coolant is used in cutting metals whereas the friction saw uses a blade and coolant is used in cutting metals. (2 + 2) (4)

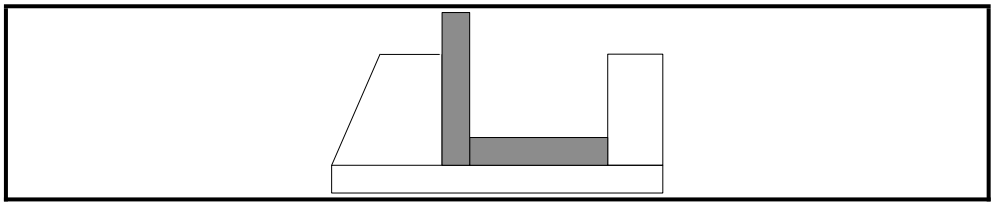
5.3 5.3.1



5.3.2



5.3.3



(3 × 2) (6)
[15]

TOTAL: 75

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